

Indicators of Energetic Status in Juvenile Coho Salmon and Chinook Salmon

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Abstract.—Bioenergetic models frequently rely on published values or models for estimating the energy density of fish, principally because of the cost and effort of obtaining direct measurements. In this study, we developed empirical models of energy density for free-ranging juvenile coho salmon *Oncorhynchus kisutch* and Chinook salmon *O. tshawytscha* sampled at sea from the west coast of Oregon to Kodiak Island, Alaska, and we evaluated the accuracy of published energy density models commonly used for these species. Our analyses showed that the energy density of juvenile coho and Chinook salmon was strongly correlated to percent dry weight and proximate constituents (especially lipid and, to a lesser extent, protein concentrations) but poorly correlated to body size and condition factor. Percent dry weight of whole fish was the single best predictor of energy density for both species, accounting for more than 90% of the variance in energy density. We also found that percent dry weight in the muscle tissue accounted for 65% of the variance in energy density. Changes in energy density mainly reflected changes in lipid composition. These results indicate that accurate estimates of energy density could be obtained at low effort and cost for juvenile coho and Chinook salmon simply by determining the water contents in whole-fish or muscle samples. Published models overestimate the energy density of juvenile coho and Chinook salmon collected from the Pacific Ocean. This may result from the extrapolation of the models to different size-classes, life stages, or habitats. More caution is needed when models are extrapolated to conditions beyond those that were used for their development.

Bioenergetic models have been used to study numerous processes, including fish growth (Hewett and Kraft 1993), the accumulation of contaminants (Trudel and Rasmussen 2001), nutrient cycling (Kraft 1992), food web dynamics (Rand and Stewart 1998), fish migration (Stockwell and Johnson 1999), and habitat selection (Brandt and Kirsch 1993), as well as for contrasting management strategies (Negus 1995). Because energy is

the currency being quantified in bioenergetic models, accurate estimates of energy densities (i.e., energy contents per unit weight) are required in these models (Hartman and Brandt 1995).

Instead of using direct measurements obtained using bomb calorimetry, bioenergetic models frequently rely on (1) published values, (2) proximate constituents, or (3) models for estimating the energy density, possibly because of the cost and time required to process a large number of samples and the difficulty of obtaining homogenates of large fish for whole-body measurements. The first approach usually assumes that the average value de-

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TABLE 1.—Published energy density models for coho salmon; ED is energy density (kJ/g [wet weight]). *W* is wet fish mass (g), and PDRY is percent dry weight.

Model	Equation	Source
M1	$ED = 5.76 + 9.86 \times 10^{-4} \cdot W$	Stewart and Ibarra (1991)
M2	$ED = 6.29 + 6.00 \times 10^{-4} \cdot W$	Rottiers and Tucker (1982)
M3	$ED = -3.25 + 0.37 \cdot PDRY$	Rottiers and Tucker (1982)

rived for a given species is constant with respect to size, season, habitats, and life stage. This assumption is inappropriate for bioenergetic models because energy density of fish can vary during ontogeny (Craig 1977; Minton and McLean 1982) and among seasons (Flath and Diana 1985; Hurst et al. 2000) and regions (Rand et al. 1994; Schultz and Conover 1997). The second approach assumes that proximate constituents can be accurately converted to energy density via energy equivalents derived for proteins and lipids. Although several energy equivalent values have been proposed for lipids and proteins (reviewed by Brett 1995), they may not always provide accurate estimates of whole-body energy (Craig et al. 1978). The third approach assumes that the energy density models are accurate. Fish energy density has generally been modeled as a function of size in bioenergetic models because size can be easily determined and is an integral component of bioenergetic models (Stewart et al. 1983; Beauchamp et al. 1989; Stewart and Ibarra 1991; Rudstam et al. 1994). These models have generally been constructed over a narrow range of size and applied by assuming that the relationship between energy density and size is stable across life stages, sizes, regions, seasons, and years. However, the relationship between energy density and size can vary among seasons, life stages, and with nutritional status (Jonsson et al. 1997; Schultz and Conover 1997; Jonsson and Jonsson 2003). Hence, alternative indicators of energy contents may be required.

Proteins and lipids are the major organic constituents in fish, whereas carbohydrates typically represent less than 2% of the whole body in fish (Higgs et al. 1995). As a consequence, the energy density of fish is expected to be primarily a function of the concentration of lipids and proteins in their tissues (Brett et al. 1969; Rottiers and Tucker 1982). Thus, surrogates for lipid and protein contents, such as condition factor or water content, could also be used to predict the energy density of fish (Hartman and Brandt 1995; Sutton et al. 2000; Jonsson and Jonsson 2003).

The objectives of this study were to examine the relationships between the energy density of ju-

venile coho salmon *Oncorhynchus kisutch* and Chinook salmon *O. tshawytscha* collected from the North Pacific as related to fish size, lipid and protein contents, condition factor, percent dry weight (i.e., 100% – percent water), and season. Because variations in energy density were expected to be primarily linked to changes in lipid concentrations (Brett et al. 1969; Rottiers and Tucker 1982), we also examined the relationships between lipid content and size, condition factor, percent dry weight, and season. We also tested models of energy density that have been published for these species. Three energy density models have been developed for Lake Michigan coho salmon (Table 1). Stewart and Ibarra (1991) derived a relationship between lipid content and size for large Lake Michigan coho salmon (1–4 kg) based on data from Reinert and Bergman (1974). They then converted lipid contents to energy density based on the relationship between energy density and lipid content obtained by Brett et al. (1969) for fingerling sockeye salmon *O. nerka* reared in the laboratory. This equation was assumed to be valid for Chinook salmon and steelhead *O. mykiss* smaller than 4 kg (Stewart and Ibarra 1991; Hanson et al. 1997). Further, Rottiers and Tucker (1982) empirically derived a relationship between energy density and fish mass and between energy density and percent dry weight for Lake Michigan coho salmon, all but one weighing 1–5 kg. Thus, the validity of these models for coho salmon smaller than 1 kg is unknown. The adequacy of these models for different but closely related species (e.g., Chinook salmon), life history stages, habitats (i.e., freshwater versus saltwater), or seasons is also unknown.

Methods

Study sites.—Juvenile coho and Chinook salmon (24–1,200 g) were collected at sea from the west coast of Oregon to Kodiak Island, Alaska (Figure 1), during the spring of 1999 and 2002, summer of 2002, fall of 1996–2002, and winter of 2001–2002 (Table 2). A rope trawl (about 90 m long $\times$ 30 m wide $\times$ 15 m deep; mesh, 6 mm; Cantrawl Pacific, Ltd., Richmond, British Columbia) was

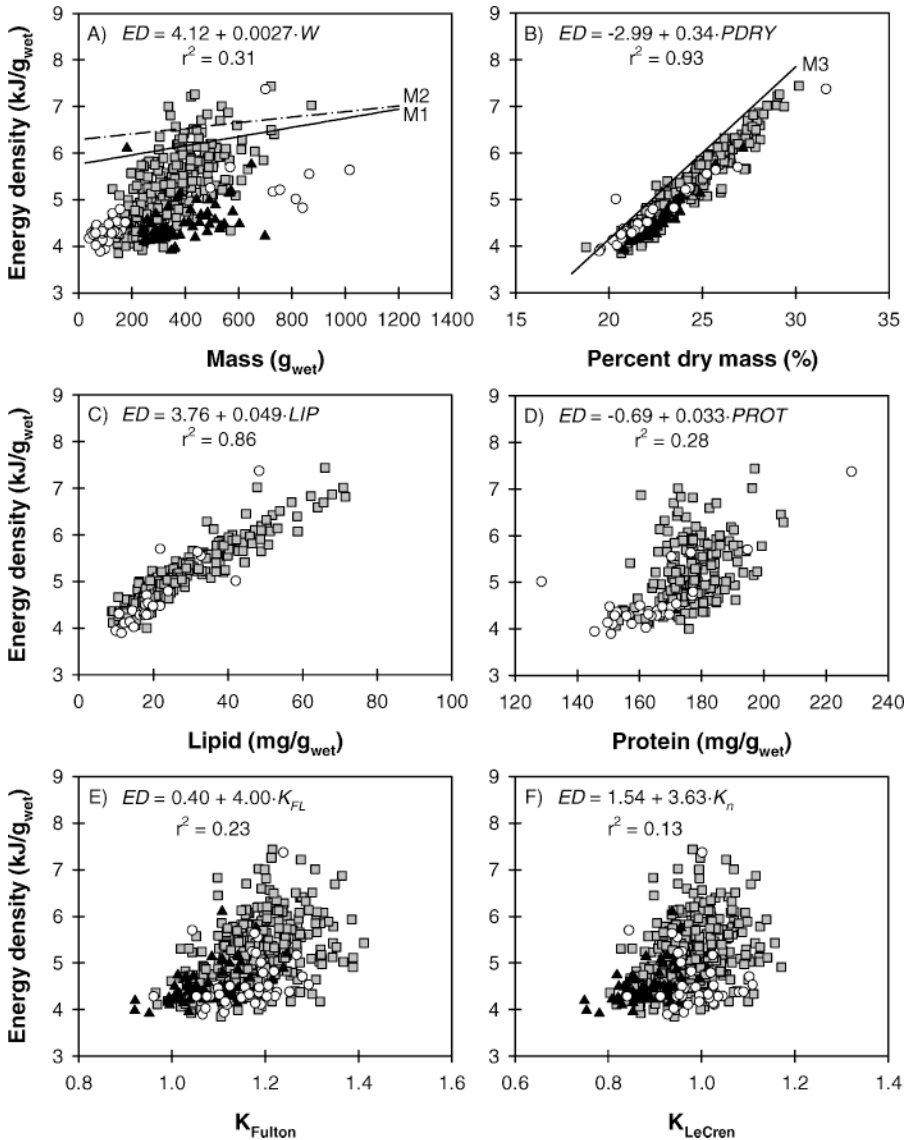


FIGURE 1.—Energy density of juvenile coho salmon in relation to (A) size (wet mass [W]), (B) percent dry mass (PDRY), (C) lipid content (LIP), (D) protein content (PROT), (E) Fulton's condition factor (K_{FL}), and (F) Le Cren's condition factor (K_n). Circles are for May–August, squares for October–November, and triangles for March; M1–M3 refer to energy density models for Lake Michigan coho salmon (Rottiers and Tucker 1982; Stewart and Ibarra 1991; see Table 1); and g_{wet} refers to 1 g of wet body mass, except on the x-axis of panel (A), where it refers to the number of grams of wet body mass.

hauled at the surface (0–20 m) for 15–30 min at 5 knots using the CCGS *W. E. Ricker*. Sampling was conducted between 0600 and 1900 hours (Pacific time). Up to 30 juvenile salmon of each species were selected from each net tow. Priority was given to fish with coded wire tags or preferred scales. The remaining fish were selected to approximately represent the size range of each spe-

cies in the catch. Fork length (± 1 mm) and mass (± 1 g) were determined on board the research vessel using a ruler and an electronic scale equipped with a counterweight to correct for ship motion. Whole fish were then frozen individually at -20°C in plastic bags for subsequent analyses.

Laboratory procedures.—Stomach contents were removed and weighed from each fish and

TABLE 2.—Number of juvenile coho and Chinook salmon included in the analysis for each sampling date; ED is energy density (kJ/g [wet weight]), and LIP is lipid content (mg/g [dry weight]).

Sampling date	Coho salmon		Chinook salmon	
	ED	LIP	ED	LIP
Oct 3–22, 1996	23	23	6	6
Nov 14–19, 1997	21	21	4	4
Oct 7–21, 1998	35	32	12	11
May 18–27, 1999	5	5	23	23
Jun 19–25, 1999	24	24	17	17
Oct 2–11, 1999	49	44	57	43
Oct 4–26, 2000	83	75	39	
Mar 10–21, 2001	27		58	
Oct 9–Nov 4, 2001	39		40	
Feb 27–Mar 17, 2002	36		59	
Jun 16–25, 2002	9		54	
Aug 15–25, 2002			37	
Oct 17–Nov 9, 2002	79		110	
Total	430	224	516	104

preserved at -20°C . Whole fish were then dried at 60°C until they reached a constant weight (7 d). Percent dry weight was determined as dry weight/wet weight $\times 100$. Dried fish were homogenized for subsequent analyses. Percent dry weight was also determined on a dorsal muscle sample (1–5 g) taken from coho and Chinook salmon caught in October 2000. Energy density based on wet weight (kJ/g) was determined using a bomb calorimeter (Phillipson 1964). Replicate analyses were performed on 10% of the samples and were generally within 0.5% (range, 0–2.9%). A subsample of the fish used for calorimetric analyses were also used for proximate analyses (Table 2). Proximate constituents (i.e., lipids and proteins) were determined using the Bligh and Dyer (1959) method and the procedures outlined by Higgs et al. (1979). Energy density was also estimated by converting proximate constituents based on assumed energy equivalents of 23.6 kJ/g for proteins and 39.5 kJ/g for lipids (Brody 1945); these estimates were compared with observed values. Carbohydrates were assumed to represent a negligible fraction of the proximate constituents in coho and Chinook salmon (Higgs et al. 1995).

Condition factors.—A plethora of condition factors have been derived for fish (Bolger and Connolly 1989; Blackwell et al. 2000). In this study, we tested three of the most commonly used indices of fish condition. First, Fulton's condition factor (K_{FL}) was calculated as

$$K_{\text{FL}} = 10^6 \cdot \frac{W}{\text{FL}^3}, \quad (1)$$

where W is fish mass (wet weight [g]) and FL is the fork length (mm). This index assumes that growth is isometric and, hence, that fish mass scales to the cube of length. However, it is difficult to interpret this index when the value of the length exponent differs from 3, because K_{FL} is correlated to size under these circumstances.

The second index used in this study, K_n or the condition factor proposed by Le Cren (1951), is independent of size. It was calculated as

$$K_n = \frac{W}{a \cdot \text{FL}^b}, \quad (2)$$

where a and b are empirically fitted constants specific for the stocks that were examined. These constants were obtained by fitting a linear regression through $\log_{10}$ transformed length and mass data from coho and Chinook salmon collected at sea from the Oregon Coast to the Bering Sea between 1995 and 2003. The relationship obtained for coho salmon was

$$\begin{aligned} \log_{10} W &= -5.19 + 3.11 \log_{10} \text{FL} \\ r^2 &= 0.99; \quad n = 3,865; \quad S_{X,Y} = 0.038 \end{aligned} \quad (3)$$

and that for Chinook salmon was

$$\begin{aligned} \log_{10} W &= -5.19 + 3.12 \log_{10} \text{FL} \\ r^2 &= 0.99; \quad n = 3,725; \quad S_{X,Y} = 0.037. \end{aligned} \quad (4)$$

These equations are valid for 90–900-mm coho salmon and 85–1,060-mm Chinook salmon. This index can only be compared within a population because the constants a and b are population specific.

The relative weight (W_r), an alternative to K_n (Anderson and Neumann 1996), is frequently used in the management of freshwater fishes (Blackwell et al. 2000). It is calculated as

$$W_r = 100 \cdot \frac{W}{W_s}, \quad (5)$$

where W_s is a species-specific model to estimate the standard weight. Because the only W_s model available for Chinook salmon was derived for inland lakes and is valid for fish larger than 200 mm (Halseth et al. 1990), we only evaluated the ability of W_r to predict energy density and lipid concentration of Chinook salmon larger than 200 mm. The relevant equation is

$$W_s = 2.18 \times 10^{-5} \cdot TL^{2.90}, \quad (6)$$

where TL is the total length (mm). We multiplied FL by 1.1 to obtain TL of Chinook salmon (M. Trudel, unpublished). Because there are currently no models to estimate W_s of coho salmon, no estimates were attempted for this species.

Statistical analyses.—The relationship between the energy density of juvenile coho and Chinook salmon and size, season, lipid, and protein contents, condition factors, and percent dry weight was examined using simple and multiple linear regressions (Sokal and Rohlf 1995). The effect of season and species were assessed using dummy variables and were included only when they were significant.

To compare model performance we used the Akaike information criterion (AIC), as generally recommended (e.g., Burnham and Anderson 1998). This approach, based on information theory, allows the selection of the most accurate competing model that has the lowest number of parameters. However, it is important to note that this index should only be used to compare models that are derived using the same data. Because proximate constituents, percent dry weight in the muscle, and the relative weight were only determined on a subset of the data collected for this study, it was not always possible to use the AIC to compare the performance of different indicators of energy status. The predictive power of nonnested models was thus simply compared using the multiple coefficient of determination (R^2). Nevertheless, to determine whether the addition of independent variables would improve nested models, we compared the relative likelihood of these functions using the AIC computed as (Burnham and Anderson 1998)

$$AIC = n \cdot \log_e(RSS/n) + 2p, \quad (7)$$

where n is the sample size, RSS is the residual sum of squares, and p is the number of parameters fitted by the regression plus one (the regression model also estimates the sum of squares). Variables that reduced the AIC by 10 units were believed to strongly support improvement in the model (Burnham and Anderson 1998).

Published energy density models were tested using two statistical procedures. First, we used a paired t -test to compare observed and predicted values. This was achieved by estimating the difference (RES_i [kJ/g]) between predicted ($PRED_i$) and observed values (OBS_i) for each i pair as

$$RES_i = PRED_i - OBS_i. \quad (8)$$

We then tested the null hypothesis that the mean difference was equal to zero using a t -test (Sokal and Rohlf 1995). A significant difference indicated the presence of a bias in the predicted values. The sign and value of the mean difference indicated the direction and magnitude of the bias, respectively. Second, we examined the relationship between deviations from observed values and the variables used to predict energy density (i.e., mass or percent dry weight) to determine the consistency of the bias. A significant correlation indicated that the bias changed with the values of the variables used to predict energy density. These analyses were performed separately for each species.

Bioenergetic model.—To evaluate the potential impacts of the biases associated with published energy density models on bioenergetic model predictions, we compared the consumption rates of juvenile coho salmon predicted by the bioenergetic model derived by Stewart and Ibarra (1991) for coho and Chinook salmon (using the energy densities predicted from their model) and those observed in our study. We assumed that coho salmon smolts migrated to sea on May 15 at 10 g and that they reached 300 g by October 15 (M. Trudel, unpublished). Prey energy density was set to 4.2 kJ/g based on the range of values reported by Davis et al. (1998) for marine prey. In addition, we used the data provided by Brett (1983) to model the seasonal cycle of water temperature.

Results

Fish Size

Energy density was 3.85–7.44 kJ/g for juvenile coho salmon and 3.55–6.74 kJ/g for Chinook salmon and increased with size (Figures 1a, 2a; Table 3). The AIC decreased by about 70–160 units when the sized-based model included a dummy variable for season, strongly indicating that the relationship between energy density and size varied among seasons (Figure 1a, 2a; Table 3). Fish collected in March had lower energy densities than those collected in the fall. In addition, the slope of the relationship between energy density and size was lower for fish caught in the fall than at the end of the winter (Figures 1a, 2a; Table 3). Overall, sized-based models had relatively low predictive power, explaining only 25–53% of the variance associated with energy density (Table 3).

Lipid concentration also increased with size in juvenile coho and Chinook salmon, though size accounted for only 17–48% of the variation in lipid contents (Figures 3a, 4a). The AIC scores of these

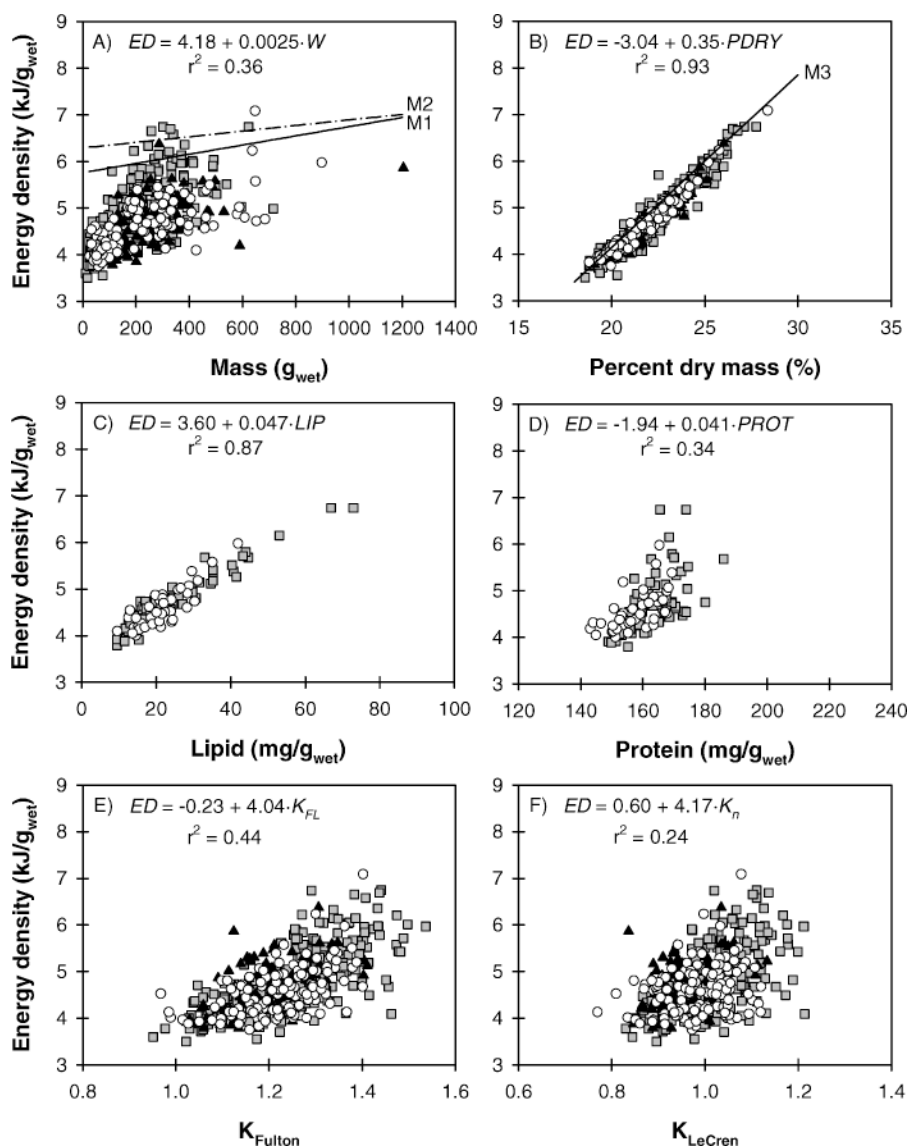


FIGURE 2.—Energy density of juvenile Chinook salmon in relation to (A) size (wet mass [W]), (B) percent dry mass (PDRY), (C) lipid content (LIP), (D) protein content (PROT), (E) Fulton's condition factor (K_{FL}), and (F) Le Cren's condition factor (K_n). See Figure 1 for additional details.

models decreased by 5–17 units when season was incorporated in these models, indicating moderate to strong evidence that the relationship between lipid concentration and size differed among seasons (Table 4).

Percent Dry Weight

Energy density increased with percent dry weight in juvenile coho and Chinook salmon (Figures 1b, 2b). Percent dry weight of whole fish explained 93% of the variance in energy density in

these fish (Figures 1b, 2b; Table 3). The relationship between energy density and percent dry weight varied significantly between species ($F_{1,941} = 286.7$, $P < 0.0001$). Including season in multiple regression analyses reduced the AIC by 44–55 units, though the predictive power of the relationship between energy density and percent dry weight was marginally improved by the addition of season (Table 3).

The energy density of the juvenile salmon caught in October 2000 also increased with percent

TABLE 3.—Empirical models of energy density for juvenile coho and Chinook salmon collected at sea from Oregon to Alaska between 1996 and 2002. Only variables that were significant at $\alpha = 0.05$ were retained in the models. Variables are defined as follows: ED, energy density (kJ/g [wet weight]); W, fish mass (g [wet weight]); FL, fork length (mm); PDRY, percent dry weight of whole fish; PDRY_{muscle}, percent dry weight of fish muscle; FA, dummy variable for the fall season (1 for fall, 0 for other seasons); SP, dummy variable for the spring season (1 for spring, 0 for other seasons); LIP, lipid content (mg/g [wet weight]); PROT, protein content (mg/g [wet weight]); n , sample size; SE, standard error of the estimate; and AIC Akaike information criterion.

Model	Equation	r^2	n	SE	AIC
Coho salmon					
A1	ED = 4.12 + 0.0027·W	0.31	430	0.60	-440
A2	ED = 4.14 + 0.0016·W + 0.0021·(W × FA)	0.53	430	0.50	-598
A3	ED = 2.64 + 0.0080·FL	0.25	430	0.63	-403
A4	ED = 2.51 + 0.0067·FL + 0.0023·(FL × FA)	0.42	430	0.55	-512
A5	ED = -2.99 + 0.34·PDRY	0.93	430	0.19	-1,402
A6	ED = -3.10 + 0.34·PDRY + 0.27·SP + 0.17·FA	0.94	430	0.18	-1,457
A7	ED = -10.85 + 0.69·PDRY _{muscle}	0.63	83	0.41	-142
A8	ED = 3.76 + 0.049·LIP	0.86	224	0.27	-585
A9	ED = -0.69 + 0.033·PROT	0.28	224	0.62	-212
A10	ED = 0.72 + 0.045·LIP + 0.018·PROT	0.94	224	0.18	-759
A11	ED = 0.40 + 4.00·K _{FL}	0.23	430	0.63	-394
A12	ED = 0.79 + 3.51·K _{FL} - 0.53·SP + 0.27·FA	0.32	430	0.60	-440
A13	ED = 1.54 + 3.63·K _n	0.13	430	0.67	-341
A14	ED = 1.59 + 3.40·K _n - 0.67·SP + 0.29·FA	0.25	430	0.62	-403
Chinook salmon					
B1	ED = 4.18 + 0.0025·W	0.36	516	0.50	-716
B2	ED = 4.14 + 0.0021·W + 0.0014·(W × FA)	0.44	516	0.47	-784
B3	ED = 3.12 + 0.0066·FL	0.36	516	0.50	-717
B4	ED = 2.98 + 0.0064·FL + 0.0014·(FL × FA)	0.42	516	0.55	-786
B5	ED = -3.04 + 0.35·PDRY	0.93	516	0.16	-1,874
B6	ED = -3.21 + 0.36·PDRY + 0.14·SP + 0.08·FA	0.94	516	0.16	-1,918
B7	ED = -12.25 + 0.78·PDRY _{muscle}	0.67	39	0.43	-61
B8	ED = 3.60 + 0.047·LIP	0.87	104	0.20	-327
B9	ED = -1.94 + 0.041·PROT	0.34	104	0.46	-158
B10	ED = 0.49 + 0.042·LIP + 0.020·PROT	0.94	104	0.14	-403
B11	ED = -0.23 + 4.03·K _{FL}	0.44	515	0.47	-781
B12	ED = -0.12 + 3.97·K _{FL} - 0.21·SP	0.45	515	0.46	-794
B13	ED = 0.60 + 4.17·K _n	0.24	515	0.54	-630
B14	ED = 0.72 + 4.08·K _n - 0.24·SP	0.27	515	0.53	-640
B15	ED = -0.21 + 0.067·W _r	0.47	428	0.44	-703

dry weight in the dorsal muscle (Table 3). This relationship also varied significantly between species ($F_{1,119} = 54.2$, $P < 0.0001$) and had lower predictive power than percent dry weight of whole fish (Table 3).

Lipid concentration increased with percent dry weight in juvenile coho and Chinook salmon (Figures 3b, 4b). Lipid concentration was also positively correlated to percent dry weight in the dorsal muscle (Table 4). As with energy density, percent dry weight was the single best predictor of lipid concentration (Table 4). In addition, the relationship between lipid concentration and percent dry weight varied significantly between species ($F_{1,325} = 41.5$, $P < 0.0001$). There was little evidence that the relationship between lipid concentration and percent dry weight differed among seasons because the AIC scores were similar with and without a dummy variable for season (Table 4). Whole

protein to lipid ratios were negatively correlated to percent dry weight ($F_{1,325} = 335.5$, $P < 0.0001$) and were significantly lower for juvenile Chinook salmon ($F_{1,325} = 34.5$, $P < 0.0001$).

Proximate Constituents

Energy density was strongly correlated to lipid content and, to a lesser extent, to protein content (Figures 1, 2). Taken together, lipids and proteins accounted for 94% of the variance in energy density for juvenile coho and Chinook salmon (Table 3). The relationship between energy density and proximate constituents did not differ significantly between juvenile coho and Chinook salmon ($F_{1,324} = 0.6$, $P > 0.4$). Thus, for both species a single equation could be used to describe the relationship between energy density and proximate constituents:

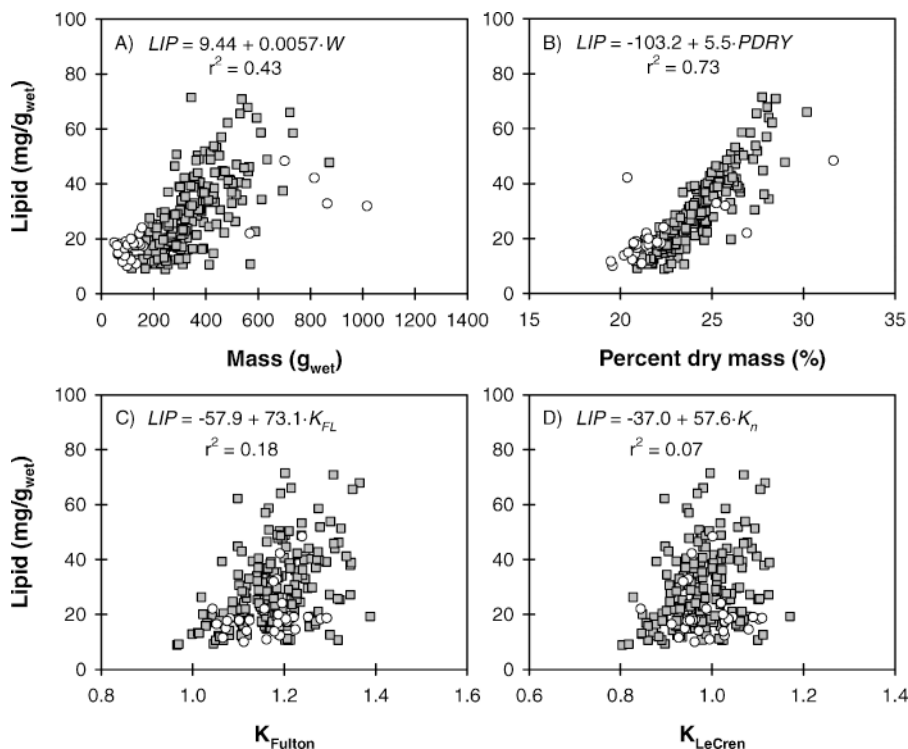


FIGURE 3.—Lipid contents of juvenile coho salmon in relation to (A) size (wet mass [W]), (B) percent dry mass (PDRY), (C) Fulton's condition factor (K_{FL}), and (D) Le Cren's condition factor (K_n). Circles are for May–August, squares for October–November; g_{wet} refers to 1 g of wet body mass, except on the x-axis of panel (A), where it refers to the number of grams of wet body mass.

$$ED = 0.75 + 0.044 \cdot LIP + 0.018 \cdot PROT, \quad (9)$$

where LIP is the lipid content (mg/g) and PROT is the protein content (mg/g). This equation indicates that lipids contribute 2.4-fold more energy than protein per gram of wet body weight.

The conversion of proximate constituents into energy units using energy equivalents for lipids and proteins generally yielded energy densities close to the observed values (Figure 5). On average, proximate constituents overestimated energy density of juvenile coho and Chinook salmon by only 84 J/g or 1.9% (Figure 5).

Condition Factor

Energy density was positively correlated to condition factor in juvenile coho and Chinook salmon; however, condition factor accounted for only 13–47% of the variance in energy density (Figures 1, 2). Adding a dummy variable for season reduced the AIC of the relationship between energy density and condition factor by 46–62 units in juvenile coho salmon and 10–13 units in juvenile Chinook salmon (Table 3). This indicated that there was

strong evidence that the relationship between energy density and condition factor varied among seasons, especially in juvenile coho salmon (Table 3).

Lipid concentration was positively correlated to condition factor in juvenile coho and Chinook salmon, though condition factor accounted for only 7–26% of the variance in lipid concentration (Figures 3, 4). With the addition of season, the relationship between lipid concentration and condition factor was improved in juvenile coho salmon but not in juvenile Chinook salmon (Table 4).

Model Evaluation

Published models of energy density that were based on size overestimated the energy density of juvenile coho and Chinook salmon by an average of 1.0–1.7 kJ/g (Figures 1a, 2a; Table 5). In addition, the bias was significantly larger than in smaller fish (Figures 1a, 2a; Table 6). The model derived by Rottiers and Tucker (1982) from energy density and percent dry weight of Lake Michigan coho salmon also systematically overestimated the

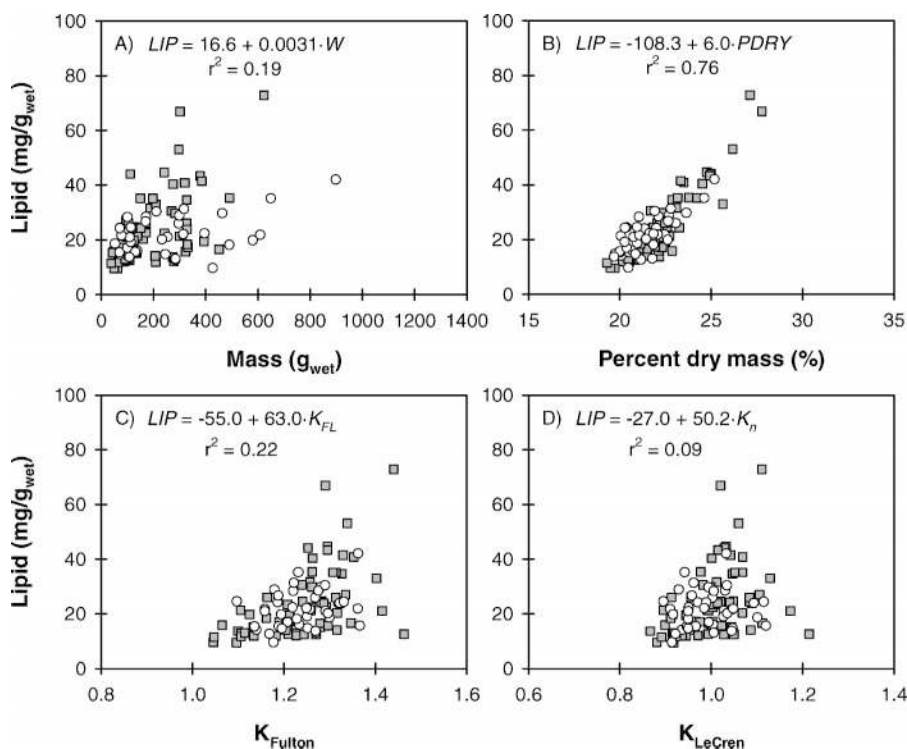


FIGURE 4.—Lipid contents of juvenile Chinook salmon in relation to (A) size (wet mass [W]), (B) percent dry mass (PDRY), (C) Fulton's condition factor (K_{FL}), and (D) Le Cren's condition factor (K_n). See Figure 3 for additional details.

TABLE 4.—Empirical models of lipid content for juvenile coho and Chinook salmon collected at sea from Oregon to Alaska between 1996 and 2000. Only variables that were significant at $\alpha = 0.05$ were retained in the models. See Table 3 for definitions of variables.

Model	Equation	r^2	n	SE	AIC
Coho salmon					
C1	$LIP = 9.44 + 0.0057 \cdot W$	0.43	224	10.4	1,064
C2	$LIP = 8.09 + 0.037 \cdot W + 0.026 \cdot (W \times FA)$	0.48	224	10.0	1,047
C3	$LIP = -23.7 + 0.18 \cdot FL$	0.38	224	10.9	1,083
C4	$LIP = 19.9 + 0.24 \cdot (FL \times FA) - 64.5 \cdot FA$	0.41	224	10.7	1,075
C5	$LIP = -103.2 + 5.5 \cdot PDRY$	0.73	224	7.2	904
C6	$LIP = -104.5 + 5.7 \cdot PDRY - 3.1 \cdot FA$	0.73	224	7.2	902
C7	$LIP = -271.1 + 13.1 \cdot PDRY_{muscle}$	0.55	75	9.4	340
C8	$LIP = -57.9 + 73.1 \cdot K_{FL}$	0.18	224	12.5	1,144
C9	$LIP = -62.1 + 70.7 \cdot K_{FL} + 8.1 \cdot FA$	0.22	224	12.3	1,136
C10	$LIP = -24.3 + 53.4 \cdot K_n$	0.07	224	13.4	1,173
C11	$LIP = -37.0 + 57.3 \cdot K_n + 10.1 \cdot FA$	0.13	224	12.9	1,160
Chinook salmon					
D1	$LIP = 16.63 + 0.0031 \cdot W$	0.19	104	10.1	485
D2	$LIP = 15.0 + 0.023 \cdot W + 0.028 \cdot (W \times FA)$	0.29	104	9.5	472
D3	$LIP = 3.31 + 0.081 \cdot FL$	0.17	104	10.2	487
D4	$LIP = 1.41 + 0.077 \cdot FL + 0.020 \cdot (FL \times FA)$	0.21	104	10.0	483
D5	$LIP = -108.3 + 6.0 \cdot PDRY$	0.76	104	5.5	358
D6	$LIP = -109.9 + 6.1 \cdot PDRY - 1.9 \cdot FA$	0.77	104	5.4	357
D7	$LIP = -55.0 + 63.0 \cdot K_{FL}$	0.22	104	9.9	480
D8	$LIP = -55.6 + 62.5 \cdot K_{FL} + 2.2 \cdot FA$	0.23	104	9.9	481
D9	$LIP = -27.0 + 50.2 \cdot K_n$	0.09	104	10.7	496
D10	$LIP = -27.1 + 49.0 \cdot K_n + 2.1 \cdot FA$	0.10	104	10.7	497
D11	$LIP = -63.0 + 1.15 \cdot W_r$	0.26	95	9.7	436

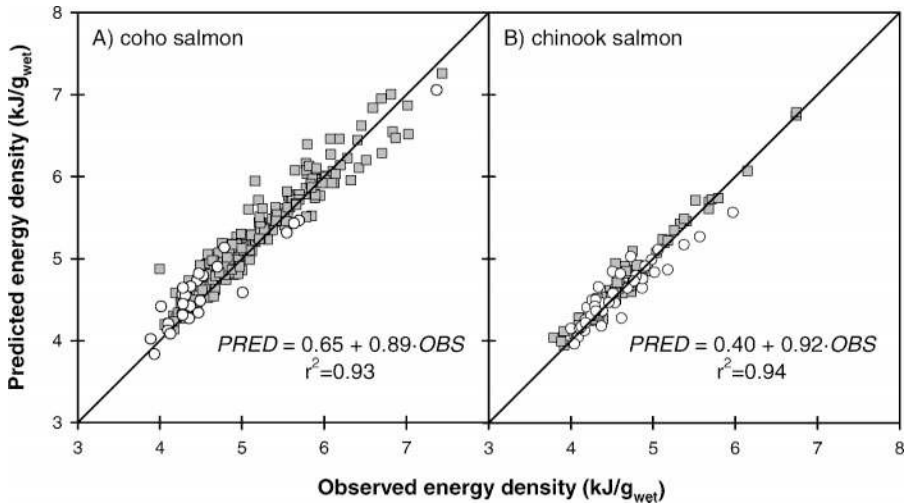


FIGURE 5.—Energy density of (A) juvenile coho salmon and (B) juvenile Chinook salmon estimated by converting proximate constituents to energy units using Brody's (1945) energy equivalents for lipids and proteins in relation to measured energy densities. The 45° line represents a 1:1 correspondence between the variables. Pearson's correlation coefficient (r^2) is provided for each comparison. Circles are for May–August, squares for October–November; g_{wet} refers to wet body mass.

energy density of juvenile coho and Chinook salmon, the bias being smaller for Chinook salmon than for coho salmon and increasing slightly with percent dry weight (Figures 1b, 2b; Tables 5, 6).

Bioenergetic Model

For coho salmon during their first growing season at sea, Stewart and Ibarra's (1991) bioenergetic model overestimated energy density by 37%, mass-specific consumption rates by 26%, and cumulative consumption rates by 17%. When the same mass-specific consumption rates were used for predicting growth, their model underestimated the final size achieved by juvenile coho salmon by over 70%.

Discussion

The analyses performed in this study showed that the energy density of juvenile coho and Chinook salmon in the North Pacific was strongly cor-

related to percent dry weight and proximate constituents (especially lipid and, to a lesser extent, protein concentrations) in whole fish, but poorly correlated to body size and condition factor. Furthermore, published models of energy density derived for Lake Michigan coho salmon overestimated the energy density of the juvenile coho and Chinook salmon we examined. Simulations indicated that bioenergetic models were quite sensitive to errors in fish energy density when they are used for predicting fish growth but less sensitive to errors in fish energy density when estimating consumption rates (Stewart et al. 1983). Thus, precision in estimates of energy density will largely depend on specific applications, but errors of the magnitude observed in this study are potentially quite serious.

Energy Density and Water Contents

Percent dry weight of whole coho and Chinook salmon was the single best predictor of their energy density, accounting for more than 90% of the variance in energy density. Strong positive correlation between energy and percent dry weight has been reported in numerous other fish species (Rottiers and Tucker 1982; Hartman and Brandt 1995; Pedersen and Hislop 2001). Although the relationship between energy density and percent dry weight varied significantly among seasons in this study, the predictive power of the regression

TABLE 5.—Mean difference ($\bar{x}$; kJ/g [wet weight]) between observed and predicted energy densities of juvenile coho and Chinook salmon.

Model	Coho salmon			Chinook salmon		
	$\bar{x}$	t (df = 429)	P	$\bar{x}$	t (df = 515)	P
M1	1.04	33.6	<0.0001	1.25	51.7	<0.0001
M2	1.45	44.7	<0.0001	1.69	67.2	<0.0001
M3	0.43	43.3	<0.0001	0.19	26.9	<0.0001

TABLE 6.—Relationship between the biases of the energy density models (M1–M3) versus the mass or percent dry weight of juvenile coho and Chinook salmon.

Comparison	Coho salmon		Chinook salmon	
	<i>r</i> (df = 428)	<i>P</i>	<i>r</i> (df = 514)	<i>P</i>
M1 versus fish mass	−0.39	<0.0001	−0.42	<0.0001
M2 versus fish mass	−0.46	<0.0001	−0.50	<0.0001
M3 versus percent dry weight	0.31	<0.0001	0.19	<0.0001

model was only improved marginally by adding a dummy variable for season. This indicates that accurate estimates of energy density in juvenile coho and Chinook salmon can be obtained at low effort and cost by simply drying the fish and determining its percent dry weight. Alternatively, water contents could be determined using total body electrical conductivity (Brown et al. 1993; Bai et al. 1994; Fisher et al. 1996). Because this approach allows the rapid determination of water contents and lean mass on live fish that are not sacrificed, it could be used to obtain repeated measures on the same individuals in the field or aquaculture facilities. However, Lantry et al. (1999) showed that total body electrical conductivity inaccurately predicted percent dry weight in five fish species, indicating that further effort is required to improve this approach before it is broadly applied.

The relationship between energy density and percent dry weight also varied between species. On average, energy density of juvenile Chinook salmon at any given percent dry weight was higher by 200 J/g than for juvenile coho salmon. This may be attributed to the observed differences in protein: lipid ratios between species. Because energy density is higher for lipids than for proteins, a dried tissue containing more lipids relative to proteins would also have higher energy density. In this study, protein to lipid ratios were lower in juvenile Chinook salmon, indicating that they had higher lipid content and hence, higher energy density in dried tissues than did juvenile coho salmon. However, differences in energy densities between these two species were relatively small (about 5%). It is important to note that this should not be taken as evidence that it is always safe to extrapolate models to closely related species. For instance, Trudel et al. (2004) recently showed that large biases were observed in oxygen consumption rate models when they were extrapolated among species of Pacific salmon. The practice of borrowing parameter estimates from other species should be carefully evaluated before it is applied (Ney 1993; Trudel et al. 2004).

This study also showed that percent dry weight

in the muscle tissue accounted for about 65% of the variance in energy density for both juvenile coho and Chinook salmon. Thus, reasonably accurate estimates of energy density could be obtained by determining percent dry weight in a muscle sample. Repeated estimates of energy density could be obtained on live fish by collecting small muscle plugs. However, the relationship derived in this study between energy density and percent dry weight in the muscle was only based on fish that were collected in the fall. This relationship may not be valid for other seasons, especially if fish have lipids stored in other locations, such as the liver and mesenteries, and do not mobilize lipids from all storage sites equally.

Energy Density and Proximate Constituents

Lipids and proteins accounted for 94% of the variance in the energy density of juvenile coho and Chinook salmon, suggesting that accurate estimates of energy density could be obtained for these species via energy equivalents derived for lipids and proteins. Several energy equivalents have been derived for lipids and proteins in fish: 17.2–23.9 kJ/g for proteins and 34.7–39.8 kJ/g for lipids (Brett 1995). This variation may reflect differences in amino acid and fatty acid composition and the percent composition of unsaturated and saturated fats (Brett 1995). The energy equivalents proposed by Brody (1945) predicted the energy density of juvenile coho and Chinook salmon reasonably well in our study. Similar results were obtained by Elliott (1976) for brown trout *Salmo trutta* reared under laboratory conditions. In contrast, Craig et al. (1978) showed that Brody's (1945) energy equivalents overestimated energy density of wild Eurasian perch *Perca fluviatilis* by 7.8%. This may have been due to taxonomic differences (i.e., salmonids versus percids), differences in lipid and protein composition, or to differences in the procedures used to determine the concentrations of proximate constituents (Iverson et al. 2001). However, these differences are relatively small and may be acceptable for some applications.

Variations in energy density are primarily linked to changes in lipid contents in fish (Brett et al. 1969; Rottiers and Tucker 1982). Similarly, energy density of juvenile coho and Chinook salmon was significantly (strongly so) correlated to lipids and, to a lesser extent to protein, concentration. However, the relative importance of lipids and protein may vary in relation to life stage, maturation, and nutritional status of fish. For instance, using the energy equivalents proposed by Brody (1945), we converted the lipid and protein concentrations measured by Parker and Vanstone (1966) for pink salmon smolts during their early sea life. For these fish, energy density was more strongly correlated to proteins ($r^2 = 0.63$) than to lipids ($r^2 = 0.40$), possibly because pink salmon smolts are rapidly growing at sea and building up muscle mass (i.e., proteins) rather than storing lipid reserves. Adult fish may also use proteins during migration and reproduction (Idler and Clemens 1959; Roff 1982), indicating that variations in energy density of the carcass could be largely influenced by changes in protein contents under these conditions.

Energy Density and Size

Energy density generally increases with size in fish, probably because lipid concentration also increases with size (Stewart et al. 1983; Van Pelt et al. 1997; Lawson et al. 1998; Hurst et al. 2000; Cochran et al. 2003). In contrast, Foy and Paul (1999) showed that the average energy density of age-0 Pacific herring *Clupea pallasii* decreased while their length and mass increased from October to March. The magnitude and direction of the correlation between energy density and fish size differed greatly among studies and species and may depend on the units used to express energy density and size (e.g., kilojoules per gram of dry versus wet weight and length versus mass), as well as the range of size used to examine the relationship, season, life stage, and nutritional status (Jonsson et al. 1997; Schultz and Conover 1997; Van Pelt et al. 1997; Pedersen and Hislop 2001; Cochran et al. 2003). In our study, energy density and lipids were poorly correlated to length and mass in juvenile coho and Chinook salmon. In addition, the slope of the relationship between energy density and mass varied among seasons. In particular, energy density tended to be lower at the end of the winter than during fall, possibly because salmon use some of the accumulated lipids to fuel their overwinter energy demands when food is less available (Beamish et al. 2004). In addition, Madenjian et al. (2000) showed that lipid concentration

was poorly correlated to size in Lake Michigan coho salmon. These results indicate that using a single relationship between energy density and size could potentially bias the outcome of bioenergetic models.

Energy Density and Condition Factor

It has been commonly assumed that condition factor can be used as an indicator of fatness, plumpness, well-being, or gonad development in fish (Blackwell et al. 2000). Several indices based on fish mass and length have been proposed over the last 100 years to describe fish condition (e.g., Fulton's condition factor, relative condition factor, and relative weight; Le Cren 1951; Bolger and Connolly 1989; Anderson and Neumann 1996). However, it is not always clear what these indices of condition represent (Liao et al. 1995). On a seasonal basis, changes in condition factor may be attributed to changes in lipids associated with the reproductive cycle or food availability (Le Cren 1951; Fechtelhelm et al. 1995). In laboratory experiments, changes in condition factor also appear to track changes in lipid contents (Brown and Murphy 1991; Thompson et al. 1991). However, field observations have produced conflicting results. Condition factor explained about 40% of the variance in the energy density of muskellunge *Esox masquinongy* and Pacific herring (Jonas et al. 1996; Paul et al. 2001). In contrast, condition factor explained about 60% of the variance in the energy density of Atlantic cod *Gadus morhua*, though the relationship was weak for values above 0.6 (Lambert and Dutil 1997). Lipids are also often poorly correlated to condition factor in free-ranging fish (Hart et al. 1940; Parker and Vanstone 1966; Gershanovich et al. 1984; Sutton et al. 2000). In our study, there was a positive correlation between energy density and condition factor and between lipid content and condition factor in juvenile coho and Chinook salmon. Condition factor accounted for only 13–47% of the variance in energy density and 7–26% of the lipid concentration variance. In addition, the relationship between energy density and condition factor and between lipid content and condition factor generally varied among seasons, indicating that condition factor is a poor indicator of fatness in juvenile coho and Chinook salmon rearing at sea. The strong positive correlation between lipid concentration and percent dry weight, coupled with the weak correlation between lipid concentration and condition factor, suggests that lipids replace water, maintaining body shape, when fat levels are increasing in fish (Hartman and

Brandt 1995). Indices of fish condition should thus be carefully evaluated with field observations rather than laboratory experiments before they can be used to assess the fatness, plumpness, or well-being of fish under field conditions.

Energy Density Models

Published models of energy density that rely on body size for predicting energy density of juvenile coho and Chinook salmon performed poorly when they were applied to our field-collected salmon. Mean bias was about 40% or 1.6 kJ/g for 50–100 g fish, tended to decrease with increasing body size for salmon caught in the fall, but remained consistently large for juvenile coho salmon caught at the end of winter and during spring.

Several factors may be responsible for the discrepancies between the observed energy densities reported in this study and the values predicted in the published models. First, these biases could have resulted from extrapolating models to smaller fish. The models we examined were developed on large coho salmon (1–5 kg) and tested on juvenile coho and Chinook salmon (mean size, 270 g). Trudel et al. (2004) showed that oxygen consumption rates of small sockeye salmon were inadequately predicted by models derived from large fish, suggesting that the parameters for size-dependent functions were not stable when they were estimated over a small range of fish sizes. The value of these allometric coefficients may also change during ontogeny (Post and Lee 1996; Bochdansky and Leggett 2001). Second, these biases may have been introduced by applying a model based on maturing fish to juveniles. Maturing fish may redistribute their stored energy for production of gametes at the expense of somatic tissues (Adams 1999). Thus, it may be inappropriate to extrapolate models to fish that are in different maturation states. Third, differences in protein: lipid ratios between Lake Michigan coho salmon and juvenile salmon rearing at sea could have produced these discrepancies. The systematic overestimation of energy density by percent dry weight, based on the model derived by Rottiers and Tucker (1982) for Lake Michigan coho salmon, suggests that juvenile coho and Chinook salmon rearing at sea have higher protein: lipid ratios than Lake Michigan coho salmon. This may reflect habitat differences and food web effects; adult Lake Michigan coho salmon are expected to be at trophic level 5 (phytoplankton → zooplankton → *Mysis* forage fish → coho salmon), compared with trophic level 4 for ocean-rearing juvenile coho and Chinook

salmon (phytoplankton → zooplankton → forage fish → coho salmon; Brodeur and Pearcy 1990; Rasmussen et al. 1990; Brodeur 1991; Vander Zanden et al. 1999). Because lipid concentration tends to increase with food chain length in fish (Rasmussen et al. 1990), Lake Michigan coho salmon would be expected to have a higher lipid content relative to protein content. Finally, for juvenile Chinook salmon, these biases might also represent species-specific differences. Regardless of the exact cause of these discrepancies, the results obtained in our study indicate that caution should always be used when models are extrapolated to conditions beyond those used in developing the model.

Management Implications

Direct measurements of energy density are generally preferable to indirect estimates. However, because of time and budget constraints, it may not always be possible to obtain direct measurements of energy density, and it may be necessary to rely on empirically derived indicators of energy density. The choice of the indicator to be used will be dictated by time and cost of measuring it, funding availability, and by the desired accuracy of the estimate.

Percent dry weight of whole fish was by far the single, best predictor of energy density in juvenile coho and Chinook salmon and can be determined relatively easily, although this approach may be impractical for very large fish. Our analyses indicate that reasonably accurate estimates of energy density could be obtained by measuring the water content in a small muscle sample (1–5 g), at least for fish collected during the fall. Further work to determine the utility of percent dry weight in muscle tissues is required to predict energy density of salmon collected in different seasons and states of nutrition.

Sized-based indices and condition factors, although easily measured, were often poorly correlated to energy density. Nevertheless, in the absence of other indicators, fish mass appeared to be the best alternative in juvenile coho salmon, at least when the season was taken into consideration. Length, mass, and Fulton's condition factor had similar predictive power for juvenile Chinook salmon when season was accounted for. The predictive power of relative weight was similar to mass and length in juvenile Chinook salmon. However, the lack of a standard weight equation for coho salmon precludes the application of the relative weight for this species. Furthermore, it may

be necessary to derive a standard weight equation for Chinook rearing at sea rather than in inland lakes.

Lastly, the energy density of juvenile coho and Chinook salmon could be accurately determined using proximate constituents, either through the regression coefficients empirically determined in this study or energy equivalents for lipids and proteins. Measuring lipids should be sufficient because it accounted for most of the variability in energy density (i.e., 86–87%). However, this approach may not be adequate for different life stages and states of nutrition. In addition, although the conversion of proximate constituents into energy units was substantially more precise than using fish size or condition factor, it is also more labor-intensive and costly than obtaining direct measurements of energy density. Nevertheless, this approach can be used for converting published values of lipids and proteins measured on free-ranging fish, provided the lipid and protein extraction procedures are similar. Furthermore, energy density can also be determined from lipids and proteins when the main objective of the work is to track changes in proximate constituents without adding any cost to the study. Finally, although energy equivalents may not always provide accurate estimates of energy density (Craig et al. 1978), they still can be used to examine relative changes in energy density over time, or among regions and species.

Energy Density and Bioenergetic Models

The bioenergetic model derived by Stewart and Ibarra (1991) for coho and Chinook salmon has generally been assumed to be reliable because Brodeur et al. (1992) successfully corroborated this model with independent estimates of growth and consumption rates. However, the close correspondence obtained between observed and predicted values by Brodeur et al. (1992) should not be taken as evidence that the model is valid in all cases. The results of our study showed that the energy densities of juvenile coho and Chinook salmon were substantially overestimated in Stewart and Ibarra's (1991) model. In addition, Trudel et al. (2004) showed that oxygen consumption rates are inadequately estimated by Stewart and Ibarra's (1991) model. Thus, although successful corroboration between independent estimates provides hope for the future, present applications should be used with caution.

Bioenergetic models provide a useful framework for testing hypotheses, illustrating concepts,

and guiding management decisions such as stocking strategies (Negus 1995). The development of bioenergetic models has proliferated quickly over the last two decades with the increasing availability of user-friendly software. Bioenergetic model parameters have been described for over 25 species (Hanson et al. 1997). However, because of difficulties in performing the laboratory and field experiments necessary to accurately estimate model parameters, investigators have alternatively extrapolated these values among species, size-classes, and life stages (Ney 1993; Trudel et al. 2004). This approach can lead to significant biases (Post 1990; Trudel et al. 2004). The findings from our study may help to refine bioenergetic model predictions by offering a low-effort, low-cost approach for estimating energy density of free-ranging salmon.

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